AI Agents Customer Support

Python (Pandas) — Data Cleaning & Data Quality Checks

Phase 1 Project Notebook

Tools: Python • Pandas • NumPy • Jupyter Notebook

Prepared from the supplied Jupyter Notebook

Notebook Overview

This document exports the notebook's markdown notes, code cells, and saved text outputs. It covers loading the Excel workbook and inspecting data tables for missing values, duplicates, data types, and shape before/while preparing the data for analysis.

Tables referenced in the notebook

• FactSupportTickets

• DimDate

• DimIssue

• DimCustomerProfile

• DimProduct

• DimSupportRoute

Code cell 1

```
import pandas as pd
import numpy as np
```

Code cell 2

```
path=pd.ExcelFile(
    r'path/to/project'
)
print(path.sheet_names)
```

Output

```
['Model Setup', 'Data Dictionary', 'FactSupportTickets', 'DimDate', 'DimIssue', 'DimCustomerProfile',
'DimProduct', 'DimSupportRoute']
```

Code cell 3

```
FactSupportTickets_df=pd.read_excel(path,sheet_name='FactSupportTickets')
```

Code cell 4

```
DimDate_df=pd.read_excel(path,sheet_name='DimDate')
```

Code cell 5

```
DimIssue_df=pd.read_excel(path,sheet_name='DimIssue')
```

Code cell 6

```
DimCustomerProfile_df=pd.read_excel(path,sheet_name='DimCustomerProfile')
```

Code cell 7

```
DimProduct_df=pd.read_excel(path,sheet_name='DimProduct')
```

Code cell 8

```
DimSupportRoute_df=pd.read_excel(path,sheet_name='DimSupportRoute')
```

Now starting Checking :- 1.Head() 2.Information (info) 3.NULL Value 4.Duplicated Value 5.shape

Starting Cleaning :----------

1. FactSupportTickets

Code cell 12

```
FactSupportTickets_df.head(10)
```

Output

```
  TicketID CreatedDate  IssueKey  ProfileKey  ProductKey  RouteKey  \
0  SUP-000001  2025-11-03         4          30           5         1
1  SUP-000002  2026-04-23         6          25           5         3
2  SUP-000003  2026-04-01        12           8           2         7
3  SUP-000004  2025-01-18         6          18           5         6
4  SUP-000005  2025-07-14         5          34           5         8
5  SUP-000006  2026-08-12         9          20           6         2
6  SUP-000007  2025-12-02         7          16           4         4
7  SUP-000008  2026-03-17         1          18           3         6
8  SUP-000009  2025-09-15         7          14           3         3
9  SUP-000010  2026-06-30         6          10           2         2

  Complexity Priority  InitialResolvedFlag  RepeatContactCount7d  ...  \
0        Low     High                    1                     0  ...
1        Low     High                    0                     1  ...
2        Low   Urgent                    1                     2  ...
3     Medium   Normal                    1                     0  ...
4     Medium   Normal                    1                     0  ...
5       High   Normal                    1                     0  ...
6        Low   Normal                    1                     0  ...
7        Low   Normal                    1                     0  ...
8        Low   Normal                    1                     0  ...
9       High   Normal                    1                     0  ...

   HumanHandlingMinutes  QueueMinutes  ResolutionHours  SLATargetHours  \
0                  0.00          0.00             0.03               8
1                 20.85         27.25            14.02              48
2                  0.00          0.00             0.03               4
3                 25.35        854.16            20.03               8
4                 26.36         20.92             1.00              24
5                 41.54         58.78            32.36              48
6                  0.00          0.00             0.07              24
7                  2.82        998.18            21.42               8
8                  4.50         44.15             1.14              48
9                 48.62         69.35            15.05              24

   SLAMetFlag  AICostEUR  HumanCostEUR  RepeatCostEUR  \
0           1       0.30          0.00           0.00
1           1       0.00         11.12          15.99
2           1       0.26          0.00          48.76
3           0       0.00         16.35           0.00
4           1       1.27         14.22           0.00
5           1       2.22         26.27           0.00
6           1       0.54          0.00           0.00
7           0       0.00          1.82           0.00
8           1       0.00          2.90           0.00
9           1       0.55         33.76           0.00

           EscalationReason     HandoffBand
0            Not applicable  Not applicable
1            Not applicable  Not applicable
2            Not applicable  Not applicable
3            Not applicable  Not applicable
4  Customer requested human     Over 12 min
5        Policy restriction     Over 12 min
6            Not applicable  Not applicable
7            Not applicable  Not applicable
8            Not applicable  Not applicable
9     Low answer confidence     Up to 5 min

[10 rows x 23 columns]
```

Code cell 13

```
FactSupportTickets_df.info()
```

Output

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 12000 entries, 0 to 11999
Data columns (total 23 columns):
 #   Column                Non-Null Count  Dtype
---  ------                --------------  -----
 0   TicketID              12000 non-null  object
 1   CreatedDate           12000 non-null  datetime64[ns]
 2   IssueKey              12000 non-null  int64
 3   ProfileKey            12000 non-null  int64
 4   ProductKey            12000 non-null  int64
 5   RouteKey              12000 non-null  int64
 6   Complexity            12000 non-null  object
 7   Priority              12000 non-null  object
 8   InitialResolvedFlag   12000 non-null  int64
 9   RepeatContactCount7d  12000 non-null  int64
 10  CSATScore             4997 non-null   float64
 11  FirstResponseMinutes  12000 non-null  float64
```

```
 12  AIConversationMinutes  12000 non-null  float64
 13  HumanHandlingMinutes   12000 non-null  float64
 14  QueueMinutes           12000 non-null  float64
 15  ResolutionHours        12000 non-null  float64
 16  SLATargetHours         12000 non-null  int64
 17  SLAMetFlag             12000 non-null  int64
 18  AICostEUR              12000 non-null  float64
 19  HumanCostEUR           12000 non-null  float64
 20  RepeatCostEUR          12000 non-null  float64
 21  EscalationReason       12000 non-null  object
 22  HandoffBand            12000 non-null  object
dtypes: datetime64[ns](1), float64(9), int64(8), object(5)
memory usage: 2.1+ MB
```

Code cell 14

```
FactSupportTickets_df['CSATScore']=FactSupportTickets_df['CSATScore'].astype('Int64')
```

Code cell 15

```
FactSupportTickets_df.info()
```

Output

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 12000 entries, 0 to 11999
Data columns (total 23 columns):
 #   Column                 Non-Null Count  Dtype
---  ------                 --------------  -----
 0   TicketID               12000 non-null  object
 1   CreatedDate            12000 non-null  datetime64[ns]
 2   IssueKey               12000 non-null  int64
 3   ProfileKey             12000 non-null  int64
 4   ProductKey             12000 non-null  int64
 5   RouteKey               12000 non-null  int64
 6   Complexity             12000 non-null  object
 7   Priority               12000 non-null  object
 8   InitialResolvedFlag    12000 non-null  int64
 9   RepeatContactCount7d   12000 non-null  int64
 10  CSATScore              4997 non-null   Int64
 11  FirstResponseMinutes   12000 non-null  float64
 12  AIConversationMinutes  12000 non-null  float64
 13  HumanHandlingMinutes   12000 non-null  float64
 14  QueueMinutes           12000 non-null  float64
 15  ResolutionHours        12000 non-null  float64
 16  SLATargetHours         12000 non-null  int64
 17  SLAMetFlag             12000 non-null  int64
 18  AICostEUR              12000 non-null  float64
 19  HumanCostEUR           12000 non-null  float64
 20  RepeatCostEUR          12000 non-null  float64
 21  EscalationReason       12000 non-null  object
 22  HandoffBand            12000 non-null  object
dtypes: Int64(1), datetime64[ns](1), float64(8), int64(8), object(5)
memory usage: 2.1+ MB
```

Code cell 16

```
FactSupportTickets_df.isna().sum()
```

Output

```
TicketID                   0
CreatedDate                0
IssueKey                   0
ProfileKey                 0
ProductKey                 0
RouteKey                   0
Complexity                 0
Priority                   0
InitialResolvedFlag        0
RepeatContactCount7d       0
CSATScore               7003
FirstResponseMinutes       0
AIConversationMinutes      0
HumanHandlingMinutes       0
QueueMinutes               0
ResolutionHours            0
SLATargetHours             0
SLAMetFlag                 0
AICostEUR                  0
HumanCostEUR               0
RepeatCostEUR              0
EscalationReason           0
HandoffBand                0
dtype: int64
```

Code cell 17

```
FactSupportTickets_df.duplicated()
```

Output

```
0        False
1        False
2        False
3        False
4        False
         ...
11995    False
11996    False
11997    False
11998    False
11999    False
Length: 12000, dtype: bool
```

Code cell 18

```
print('Column',FactSupportTickets_df.shape[0])
print('Row',FactSupportTickets_df.shape[1])
```

Output

```
Column 12000
Row 23
```

2.DimDate

Code cell 20

```
DimDate_df.head(10)
```

Output

```
Date  Year Quarter  MonthNo MonthName YearMonth  YearMonthSort
0 2025-01-01  2025      Q1        1       Jan   2025-01         202501
1 2025-01-02  2025      Q1        1       Jan   2025-01         202501
2 2025-01-03  2025      Q1        1       Jan   2025-01         202501
3 2025-01-04  2025      Q1        1       Jan   2025-01         202501
4 2025-01-05  2025      Q1        1       Jan   2025-01         202501
5 2025-01-06  2025      Q1        1       Jan   2025-01         202501
6 2025-01-07  2025      Q1        1       Jan   2025-01         202501
7 2025-01-08  2025      Q1        1       Jan   2025-01         202501
8 2025-01-09  2025      Q1        1       Jan   2025-01         202501
9 2025-01-10  2025      Q1        1       Jan   2025-01         202501
```

Code cell 21

```
DimDate_df.info()
```

Output

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 730 entries, 0 to 729
Data columns (total 7 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   Date           730 non-null    datetime64[ns]
 1   Year           730 non-null    int64
 2   Quarter        730 non-null    object
 3   MonthNo        730 non-null    int64
 4   MonthName      730 non-null    object
 5   YearMonth      730 non-null    object
 6   YearMonthSort  730 non-null    int64
dtypes: datetime64[ns](1), int64(3), object(3)
memory usage: 40.1+ KB
```

Code cell 22

```
# correction of data type YearMonth
DimDate_df['YearMonth'] = pd.to_datetime(
    DimDate_df['YearMonth'],
    format='%Y-%m',
    errors='coerce'
)
```

Code cell 23

```
DimDate_df.isna().sum()
```

Output

```
Date            0
Year            0
Quarter         0
MonthNo         0
MonthName       0
YearMonth       0
YearMonthSort   0
dtype: int64
```

Code cell 24

```
DimDate_df.duplicated()
```

Output

```
0      False
1      False
2      False
3      False
4      False
       ...
725    False
726    False
727    False
728    False
729    False
Length: 730, dtype: bool
```

Code cell 25

```
print('Row',DimDate_df.shape[1])
print('Column',DimDate_df.shape[0])
```

Output

```
Row 7
Column 730
```

3.Dim Issue

Code cell 27

```
DimIssue_df.head(10)
```

Output

IssueKey	IssueCategory	IssueSubcategory	SupportTeam
0	1 Account access	Password reset	Customer care
1	2 Account access	Account locked	Customer care
2	3 Account access	Identity verification	Customer care
3	4 Orders & delivery	Order tracking	Customer care
4	5 Orders & delivery	Delivery delay	Customer care
5	6 Orders & delivery	Missing parcel	Customer care
6	7 Billing & refunds	Invoice copy	Billing care
7	8 Billing & refunds	Refund eligibility	Billing care
8	9 Billing & refunds	Billing dispute	Billing care
9	10 Technical support	Setup guidance	Technical care

Code cell 28

```
DimIssue_df.info()
```

Output

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 15 entries, 0 to 14
Data columns (total 4 columns):
 #   Column            Non-Null Count  Dtype 
---  ------            --------------  ----- 
 0   IssueKey          15 non-null     int64 
 1   IssueCategory     15 non-null     object
 2   IssueSubcategory  15 non-null     object
 3   SupportTeam       15 non-null     object
dtypes: int64(1), object(3)
memory usage: 612.0+ bytes
```

Code cell 29

```
DimIssue_df.isna().sum()
```

Output

```
IssueKey           0
IssueCategory      0
IssueSubcategory   0
```

```
SupportTeam         0
dtype: int64
```

Code cell 30

```
DimIssue_df.duplicated()
```

Output

```
0     False
1     False
2     False
3     False
4     False
5     False
6     False
7     False
8     False
9     False
10    False
11    False
12    False
13    False
14    False
dtype: bool
```

Code cell 31

```
print('Row',DimIssue_df.shape[1])
print('Column',DimIssue_df.shape[0])
```

Output

```
Row 4
Column 15
```

4. Dim Customer Profile

Code cell 33

```
DimCustomerProfile_df.head(10)
```

Output

```
ProfileKey           Region         Country Language CustomerTier  \
0           1  Northern Europe  United Kingdom  English     Standard
1           2  Northern Europe  United Kingdom  English     Standard
2           3  Northern Europe  United Kingdom  English         Plus
3           4  Northern Europe  United Kingdom  English         Plus
4           5  Northern Europe  United Kingdom  English     Business
5           6  Northern Europe  United Kingdom  English     Business
6           7  Northern Europe          Sweden  Swedish     Standard
7           8  Northern Europe          Sweden  Swedish     Standard
8           9  Northern Europe          Sweden  Swedish         Plus
9          10  Northern Europe          Sweden  Swedish         Plus

     Lifecycle
0          New
1  Established
2          New
3  Established
4          New
5  Established
6          New
7  Established
8          New
9  Established
```

Code cell 34

```
DimCustomerProfile_df.info()
```

Output

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 36 entries, 0 to 35
Data columns (total 6 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   ProfileKey    36 non-null     int64
 1   Region        36 non-null     object
 2   Country       36 non-null     object
 3   Language      36 non-null     object
 4   CustomerTier  36 non-null     object
 5   Lifecycle     36 non-null     object
dtypes: int64(1), object(5)
```

```
memory usage: 1.8+ KB
```

Code cell 35

```
DimCustomerProfile_df.isna().sum()
```

Output

```
ProfileKey      0
Region          0
Country         0
Language        0
CustomerTier    0
Lifecycle       0
dtype: int64
```

Code cell 36

```
DimCustomerProfile_df.duplicated().head(10)
```

Output

```
0    False
1    False
2    False
3    False
4    False
5    False
6    False
7    False
8    False
9    False
dtype: bool
```

Code cell 37

```
print('Row',DimCustomerProfile_df.shape[1])
print('Column',DimCustomerProfile_df.shape[0])
```

Output

```
Row 6
Column 36
```

5. Dim Product

Code cell 39

```
DimProduct_df.head(10)
```

Output

```
ProductKey ProductFamily            ProductName
0           1      Commerce           Shop platform
1           2      Commerce  Marketplace connector
2           3  Productivity          Team workspace
3           4  Productivity      Automation studio
4           5       Devices              Smart hub
5           6       Devices             Home sensor
```

Code cell 40

```
DimProduct_df.info()
```

Output

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 6 entries, 0 to 5
Data columns (total 3 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   ProductKey     6 non-null      int64
 1   ProductFamily  6 non-null      object
 2   ProductName    6 non-null      object
dtypes: int64(1), object(2)
memory usage: 276.0+ bytes
```

Code cell 41

```
DimProduct_df.isna().sum()
```

Output

```
ProductKey       0
ProductFamily    0
```

```
ProductName      0
dtype: int64
```

Code cell 42

```
DimProduct_df.duplicated()
```

Output

```
0    False
1    False
2    False
3    False
4    False
5    False
dtype: bool
```

Code cell 43

```
print('Row',DimProduct_df.shape[1])
print('Column',DimProduct_df.shape[0])
```

Output

```
Row 3
Column 6
```

6. Dim Support Route

Code cell 45

```
DimSupportRoute_df.head(10)
```

Output

```
RouteKey    Channel SupportPath    AIAgent PathColor
0         1  Live chat     AI only  Assist AI   #007F82
1         2  Live chat  AI → human  Assist AI   #D68A32
2         3  Live chat  Human only      No AI   #4977C8
3         4      Email     AI only  Assist AI   #007F82
4         5      Email  AI → human  Assist AI   #D68A32
5         6      Email  Human only      No AI   #4977C8
6         7  Messaging     AI only  Assist AI   #007F82
7         8  Messaging  AI → human  Assist AI   #D68A32
8         9  Messaging  Human only      No AI   #4977C8
```

Code cell 46

```
DimSupportRoute_df.info()
```

Output

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9 entries, 0 to 8
Data columns (total 5 columns):
 #   Column       Non-Null Count  Dtype 
---  ------       --------------  ----- 
 0   RouteKey     9 non-null      int64 
 1   Channel      9 non-null      object
 2   SupportPath  9 non-null      object
 3   AIAgent      9 non-null      object
 4   PathColor    9 non-null      object
dtypes: int64(1), object(4)
memory usage: 492.0+ bytes
```

Code cell 47

```
DimSupportRoute_df.isna().sum()
```

Output

```
RouteKey       0
Channel        0
SupportPath    0
AIAgent        0
PathColor      0
dtype: int64
```

Code cell 48

```
DimSupportRoute_df.duplicated()
```

Output

```
0    False
```

```
1    False
2    False
3    False
4    False
5    False
6    False
7    False
8    False
dtype: bool
```

Code cell 49

```
print('Row',DimSupportRoute_df.shape[1])
print('Column',DimSupportRoute_df.shape[0])
```

Output

```
Row 5
Column 9
```

Code cell 50

```
from sqlalchemy import create_engine

username = "postgres"
password = "8540"
database = "AI & Agents"
host = "localhost"
port = 5432

engine = create_engine(
    f"postgresql+psycopg2://{username}:{password}@{host}:{port}/{database}"
)
```

Code cell 51

```
FactSupportTickets_df.to_sql(
    "FactSupportTickets_df",
    engine,
    if_exists="replace",
    index=False
)
print("FactSupportTickets_df connected PostgreSQL")
```

Output

```
FactSupportTickets_df connected PostgreSQL
```

Code cell 52

```
DimDate_df.to_sql(
    "DimDate_df",
    engine,
    if_exists="replace",
    index=False
)
print("DimDate_df Connected PostgresSQL")
```

Output

```
DimDate_df Connected PostgresSQL
```

Code cell 53

```
DimIssue_df.to_sql(
    "DimIssue_df",
    engine,
    if_exists="replace",
    index=False
)
print("DimIssue_df Connected PostgresSQL")
```

Output

```
DimIssue_df Connected PostgresSQL
```

Code cell 54

```
DimCustomerProfile_df.to_sql(
    "DimCustomerProfile_df",
    engine,
    if_exists="replace",
    index=False
)
print("DimCustomerProfile_df Connected PostgresSQL")
```

Output

```
DimCustomerProfile_df Connected PostgresSQL
```

Code cell 55

```
DimProduct_df.to_sql(
    "DimProduct_df",
    engine,
    if_exists="replace",
    index=False
)
print("DimProduct_df Connected PostgresSQL")
```

Output

```
DimProduct_df Connected PostgresSQL
```

Code cell 56

```
DimSupportRoute_df.to_sql(
    "DimSupportRoute_df",
    engine,
    if_exists="replace",
    index=False
)
print("DimSupportRoute_df Connected PostgresSQL")
```

Output

```
DimSupportRoute_df Connected PostgresSQL
```